ADITYA MALVIYA

Senior Data Engineer | Azure Data Engineer | Azure Databricks, PySpark, Azure Data Factory, Delta Lake

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PROFESSIONAL SUMMARY

Senior Data Engineer with nearly 5 years of experience at Tata Consultancy Services (TCS), building production Azure data platforms for Rolls-Royce. Designs batch and real-time ETL/ELT pipelines processing ~27 GB/day with Azure Databricks, PySpark, Spark SQL, Azure Data Factory (ADF), ADLS Gen2, Delta Lake and Unity Catalog on a Medallion (Lakehouse) Architecture. Cut Spark job runtime by 73% and Azure costs by ~40%, and leads a 6-member team as Scrum Master. Microsoft Certified: Fabric Data Engineer Associate (DP-700).

TECHNICAL SKILLS

Languages: Python, SQL, T-SQL, PySpark, Spark SQL

Big Data & Processing: Apache Spark, Spark Structured Streaming, Apache Kafka, Spark Performance Tuning, Adaptive Query Execution (AQE), Data Skew Handling

Azure Data Platform: Azure Databricks, Azure Data Factory (ADF), Azure Data Lake Storage Gen2 (ADLS Gen2), Azure Event Hubs, Azure Blob Storage, Azure Key Vault, Azure DevOps, Azure API Management, Microsoft Fabric

Databricks & Lakehouse: Delta Lake, Unity Catalog, Databricks Workflows, Auto Loader, Delta MERGE, OPTIMIZE / Z-ORDER, Lakehouse Architecture, Medallion Architecture (Bronze, Silver, Gold)

Data Modeling & Warehousing: ETL, ELT, Data Pipelines, Data Warehousing, Dimensional Modeling, Star Schema, Slowly Changing Dimensions (SCD Type 2), Incremental Loads, Change Data Capture (CDC)

Data Governance & Quality: Data Quality Checks, Data Governance, Data Lineage, Role-Based Access Control (RBAC)

Databases & BI: Microsoft SQL Server, PostgreSQL, MySQL, Redis, DuckDB, Power BI

DevOps & Engineering: CI/CD (Azure DevOps Pipelines, YAML), Git, GitHub Actions, Docker, Pytest, SonarQube, FastAPI, REST APIs, dbt

Cloud (other): Google Cloud Platform (GCP), Databricks on GCP, Vertex AI, Google Cloud Functions

Methodologies & Leadership: Agile, Scrum Master, Sprint Planning, Code Reviews, Team Leadership, Mentoring, Stakeholder Management

Working Knowledge: Azure Synapse Analytics, Delta Live Tables (DLT), Apache Airflow

PROFESSIONAL EXPERIENCE

Senior Data Engineer | Tata Consultancy Services (TCS) | Mumbai, India | Dec 2021 – Present

Promoted from Data Engineer to Senior Data Engineer in Mar 2026. Clients: Rolls-Royce Digital (aerospace) and Avery Dennison (US Fortune 500).

Rolls-Royce Digital – Engine History Analysis, Azure Lakehouse Platform

- Design and own production Azure Data Factory (ADF) and Azure Databricks ETL/ELT pipelines in PySpark and Spark SQL on a Medallion Architecture over Delta Lake and ADLS Gen2, processing ~27 GB/day streaming and 5–7 GB/day batch data into star-schema models for Power BI.
- Built real-time ingestion with Azure Event Hubs, Apache Kafka and Spark Structured Streaming, plus incremental loads using Delta MERGE, SCD Type 2 and Auto Loader, orchestrated with Databricks Workflows.
- Cut a critical Spark job's runtime by 73% (52 to 14 minutes) by fixing join skew with key salting and Adaptive Query Execution (AQE), eliminating executor out-of-memory failures.
- Reduced Azure costs by ~40% through Databricks cluster right-sizing, Delta OPTIMIZE, Z-ORDER, partitioning and parameterized ADF pipelines.
- Implemented layer-wise data quality checks and Unity Catalog governance (RBAC, data lineage); secured credentials with Azure Key Vault-backed secret scopes and service principals.
- Promote ADF and Databricks code across dev, test and production with Azure DevOps CI/CD pipelines (YAML, Git branching).
- Lead a 6-member offshore team as Scrum Master: run stand-ups and sprint planning, manage Azure DevOps boards, mentor juniors, review pull requests and own delivery commitments to the client.

Rolls-Royce Digital – Assumptions Management, Data Services and APIs

- Built production FastAPI data services with RBAC, Azure API Management and Docker over SQL Server and Blob Storage; improved performance by ~30% with Redis caching and SQL/T-SQL query optimization.
- Raised code quality by resolving SonarQube findings and adding Pytest unit tests in Agile/Scrum sprints.

Avery Dennison (Smartrac RFID) – R&D Data and AI Platform

- Built Databricks pipelines on Google Cloud Platform (GCP) to ingest, cleanse and prepare RFID R&D datasets for training machine learning models behind an AI research tool now in production.
- Developed backend services with Vertex AI, Google Cloud Functions and a vector database for semantic search.

Python Developer Intern | BYJU'S (Gradeup) | Remote | Jul 2021 – Nov 2021

- Built Python scripts and REST API integrations for Gradeup, an exam-preparation platform acquired by BYJU'S.

FREELANCE WORK

Moriveda – D2C E-commerce Platform | moriveda.in **| May 2026 – Jun 2026**

- Built and launched a live e-commerce site with a Python/PHP backend and SQL database, integrating Razorpay payments and iThinkLogistics shipping APIs.
- Automated scheduled jobs for payment reconciliation, abandoned-cart recovery and review requests using cron.

PROJECTS

- **Corpus – Document AI and Data Q&A Workspace (Oct 2026 – Present):** Browser-native RAG app with hybrid BM25 + vector retrieval, intent routing and sentence-level citations, running Gemma 4 on WebGPU; turns questions about CSV/Parquet files into guarded DuckDB SQL and scores retrieval with hit@k and MRR. Live | GitHub
- **Orchestra – Pipeline Orchestrator and Run Doctor (Oct 2026 – Present):** Airflow-style scheduler on a simulated clock with cron, catch-up, retries with exponential backoff, pools, trigger rules, SLAs and idempotent backfills; diagnoses failures (root cause vs blast radius) and exports DAGs to Apache Airflow 2. Live | GitHub
- **FlowForge – Visual ETL Pipeline Builder (Oct 2026 – Present):** Drag-and-drop ETL designer that generates DuckDB SQL, Spark SQL, PySpark and dbt code with data-quality tests. Live | GitHub
- **Research: AQE vs Salting for Spark Skewed Joins (Aug 2026):** Reproducible benchmarks and cost models comparing Adaptive Query Execution and selective salting. GitHub
- **PipeScope – Spark Performance and Lineage Analyzer (Aug 2025 – Present):** Analyzes Spark event logs for tuning issues and traces SQL/dbt column-level lineage. Live | GitHub
- **DataLens – Data Profiling and Quality Tool (Jul 2024 – Present):** Profiles CSV, Parquet and JSON files with DuckDB-WASM and scores data completeness, validity and uniqueness. Live | GitHub
- **QueryRoom – SQL Practice Studio (Apr 2023 – Present):** In-browser SQLite workspace with 332 SQL interview questions, query execution and answer verification. Live | GitHub

CERTIFICATIONS

- Microsoft Certified: Fabric Data Engineer Associate (DP-700)
- Microsoft Certified: Azure Data Fundamentals (DP-900)

EDUCATION

Bachelor of Technology (B.Tech), Mechanical Engineering | Bhilai Institute of Technology, Durg | 2021
Grade: 71%